DISK-LIKE SELF-SIMILAR TILES WITH HIGH BOUNDARY DIMENSION

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Abstract. We construct an infinite family of self-similar plane tiles $\{A_n\}_{n\geq 3}$, each generated by n contractions of ratio $1/\sqrt{n}$, with the following properties: (i) the neighbor graph has exactly 6 boundary components for every n; (ii) each A_n is disk-like (homeomorphic to a closed disk); (iii) the boundary Hausdorff dimension satisfies $2 - \dim_{\mathrm{H}}(\partial A_n) \sim 8/(n^2 \ln n)$ as $n \to \infty$. In particular, $\dim_{\mathrm{H}}(\partial A_n) \to 2$ with bounded neighbor graph complexity. This complements the examples of Keesling [6], where $\dim_{\mathrm{H}}(\partial A) \to n$ in $\mathbb{R}^n$ but the graph complexity grows without bound.

1. Introduction

A *self-similar tile* is a compact set $A \subset \mathbb{R}^2$ with non-empty interior satisfying

$$A = \bigcup_{i=0}^{N-1} f_i(A),$$

where the f_i are contractive similarities [5] and the union is measure-disjoint (the *Open Set Condition*, hereafter OSC). Such tiles arise naturally in the study of self-affine lattice tilings [7]. The *boundary* ∂A is a compact fractal set whose Hausdorff dimension $s = \dim_{\mathrm{H}}(\partial A)$ satisfies $1 \leq s < 2$ (see [4] for background on Hausdorff dimension).

Achieving $\dim_{\mathrm{H}}(\partial A)$ close to 2 is easy in principle. Keesling [6] proved that $\dim_{\mathrm{H}}(\partial A) < n$ for every self-similar tile in $\mathbb{R}^n$ and gave examples showing that $\dim_{\mathrm{H}}(\partial A)$ can be arbitrarily close to n (see also Duvall, Keesling and Strichartz [3] for explicit calculations of boundary dimensions). In those examples the IFS has $4k+3$ maps on $\mathbb{R}$ with every fourth digit offset, and the number of boundary components (the *neighbor graph complexity*) grows without bound as $k \to \infty$.

The disk-like constraint is much more restrictive. High boundary dimension corresponds to extreme geometric roughness of ∂A, and tiles with large $\dim_{\mathrm{H}}(\partial A)$ are generically non-disk-like (e.g., the Lévy C curve with $\dim_{\mathrm{H}}(\partial A) \approx 1.934$ is not a topological disk). Achieving $\dim_{\mathrm{H}}(\partial A)$ close to 2 while maintaining the disk-like property and bounded graph complexity is therefore a much harder problem.

The natural question arises: *can* $\dim_{\mathrm{H}}(\partial A)$ *approach* 2 *with a bounded number of boundary components and the disk-like property?*

Main result. We construct an infinite family of self-similar tiles $\{A_n\}_{n\geq 3}$ with exactly 6 boundary components for every $n \geq 3$, and $\dim_{\mathrm{H}}(\partial A_n) \to 2$ as $n \to \infty$. Each tile is *disk-like* (homeomorphic to a closed disk). Furthermore, the rate of convergence is sharp:

$$2 - \dim_{\mathrm{H}}(\partial A_n) \sim \frac{8}{n^2 \ln n}.$$

The boundary dimension is determined by a quartic polynomial $p_n(x)$ with integer coefficients depending on n, for which we derive a closed-form expression. Both the constancy of the neighbor graph (always 6 components) and the disk-like property are proved for all $n \geq 3$ by an explicit analysis of the refinement equation.

2. The Comb Family

2.1. **Definition via companion matrix.** For an integer $n \geq 3$, consider the companion matrix of the polynomial $x^2 - n$:

$$g = \begin{pmatrix} 0 & n \\ 1 & 0 \end{pmatrix}, \tag{1}$$

with eigenvalues $\pm\sqrt{n}$ and spectral radius $\sqrt{n}$.

Definition 1 (Comb IFS)**.** *For $n \geq 3$, the* comb IFS with parameter n *consists of n digit maps acting on $\mathbb{R}^2$:*

$$\begin{aligned} h_j(x) &= x + \big(j(n-1),\, 0\big), \qquad & j = 0, 1, \ldots, n-2, \\ h_{n-1}(x) &= -x + \big(n^2-n-1,\; 2n-3\big), \end{aligned} \tag{2}$$

with attractor

$$A_n = g^{-1}\big(h_0(A_n) \cup h_1(A_n) \cup \cdots \cup h_{n-1}(A_n)\big). \tag{3}$$

All digits are integer vectors. The map h_{n-1} includes a central reflection $x \mapsto -x$.

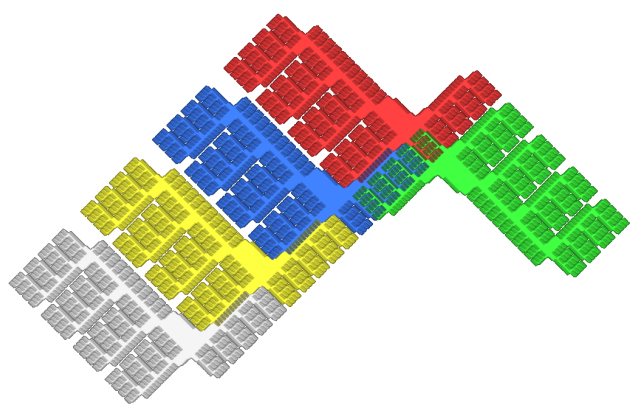

Figure 1. The comb tile A_5 ($n = 5$, five maps). Each first-level piece $f_j(A_5)$ is shown in a distinct colour: the four spine pieces $f_0, \ldots, f_3$ tile the lower-left staircase, while the single tooth piece f_4 (with central reflection) extends to the upper right.

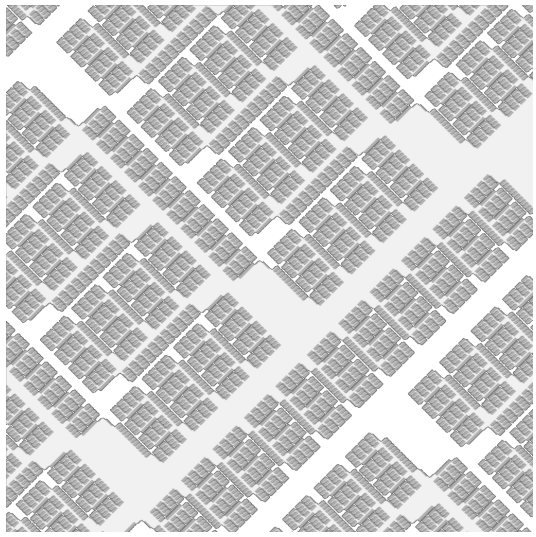

FIGURE 2. Magnified fragment of the boundary ∂A_5. The boundary is a Jordan curve (A_5 is disk-like by Theorem 11), yet its Hausdorff dimension is $\dim_{\mathrm{H}}(\partial A_5) \approx 1.863$ — visibly close to space-filling.

Each contraction $f_j = g^{-1} \circ h_j$ is an affine similarity with linear part g^{-1} (for $j < n-1$) or $-g^{-1}$ (for $j = n-1$), and common contraction ratio $1/\sqrt{n}$.

Remark 2 (Digit set geometry). The first $n-1$ digits $h_0, \ldots, h_{n-2}$ are pure translations along the first coordinate axis, with displacements 0, $n-1$, $2(n-1)$, $\ldots$, $(n-2)(n-1)$. These form the *spine* of the comb. The single map h_{n-1}, which includes a central reflection, has an off-axis translation component and forms the *tooth* of the comb. This asymmetric digit geometry — one outlier map among $n-1$ collinear ones — is the structural source of the small neighbor graph size (see Section 3). The name *comb* reflects the first-level subdivision visible in Figure 1: the $n-1$ spine pieces form a row of teeth, while the single reflected piece extends sideways like a handle.

2.2. **Basic properties.** Since all n maps share the contraction ratio $r = 1/\sqrt{n}$, the Hausdorff dimension of the attractor is

$$\dim_{\mathrm{H}}(A_n) = \frac{\log n}{\log \sqrt{n}} = 2$$

for every n. Thus each A_n is a plane-filling set (a tile).

Proposition 3 (Self-replicating tiling)**.** *For every $n \geq 3$, A_n is an n-rep tile in the sense of* [1]*: the plane is tiled by copies of A_n and its half-turn $-A_n$:*

$$\mathbb{R}^2 = \bigcup_{v \in \mathbb{Z}^2} (A_n + v) \;\cup\; \bigcup_{w \in \mathbb{Z}^2} (-A_n + w),$$

with the union measure-disjoint.

The proof follows from the neighbor graph analysis (Theorem 8), which verifies the Open Set Condition with $V = \operatorname{int} A_n$ and shows that every neighbor of A_n is either a $\mathbb{Z}^2$-translate or a half-turn $\mathbb{Z}^2$-translate [1].

Following [1], the maps above are written in $\mathbb{R}^2$ with the integer lattice $\mathbb{Z}^2$, so that g and all digits have integer entries. Since g has two real eigenvalues $\pm\sqrt{n}$ with independent eigenvectors, one can choose a basis in which g acts as a Euclidean similarity, making the IFS self-similar (rather than merely self-affine).

Remark 4 (Complex form of the IFS). Identify the eigenplane of g with $\mathbb{C}$ by mapping the eigenvectors for $\sqrt{n}$ and $-\sqrt{n}$ to the real and imaginary axes, respectively. Then the expansion acts as

$$g(z) = \sqrt{n}\,\bar{z},$$

an anti-conformal map of modulus $\sqrt{n}$. Since $g^{-1}(z) = \bar{z}/\sqrt{n}$, the IFS takes the form

$$\begin{aligned} f_j(z) &= \frac{\bar{z}}{\sqrt{n}} + c_j, \qquad\qquad j = 0, \dots, n{-}2, \\ f_{n-1}(z) &= -\frac{\bar{z}}{\sqrt{n}} + c_{n-1}, \end{aligned} \tag{4}$$

where the spine translations are uniformly spaced along $\mathbb{R}\cdot(1{+}i)$:

$$c_j = \frac{j(n{-}1)(1{+}i)}{2n}, \qquad j = 0, \dots, n{-}2,$$

and the tooth translation is

$$c_{n-1} = \frac{(n^2{-}n{-}1)(1{+}i) + (2n{-}3)\sqrt{n}\,(1{-}i)}{2n}.$$

All maps are anti-conformal similarities of ratio $1/\sqrt{n}$. The spine digits are rational, but the tooth digit involves $\sqrt{n}$. The rational-space representation of Definition 1, where *all* digits are integer vectors, is therefore preferred for the exact neighbor graph analysis in Section 3.

Remark 5 (The case $n = 2$). For $n = 2$ the attractor A_2 is a rectangle (a non-fractal, disk-like tile). The quartic (11) gives $p_2(x) = x(x{-}1)(x^2{-}2)$ with dominant root $\sqrt{2}$ and $\dim_{\mathrm{H}}(\partial A_2) = 1$, correctly matching the boundary dimension. However, the neighbor graph has 8 components (not 6), so the specific graph structure and substitution matrix derived below apply only to $n \geq 3$.

3. The Neighbor Graph

In this section we derive the neighbor graph of the comb IFS from first principles, proving that it has exactly 6 boundary components for every $n \geq 3$.

3.1. **Neighbor map algebra.** A *neighbor map* of the tile $A = A_n$ is an isometry $\sigma\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $A \cap \sigma(A) \neq \emptyset$ and $\sigma \neq \mathrm{id}$ [1]. Because every contraction $f_j = g^{-1} \circ h_j$ has linear part $\pm g^{-1}$, every neighbor map has the form $\sigma(x) = \varepsilon x + t$ with $\varepsilon \in \{+I, -I\}$ and $t \in \mathbb{R}^2$.

The *refinement rule*: substituting $A = \bigcup_i f_i(A)$ into both sides of $A \cap \sigma(A)$ gives

$$A \cap \sigma(A) = \bigcup_{i,j} f_i\big(A \cap \sigma'_{ij}(A)\big), \qquad \sigma'_{ij} = f_i^{-1} \circ \sigma \circ f_j.$$

A direct calculation yields

$$\sigma'_{ij}(x) = \varepsilon_i \varepsilon \varepsilon_j\, x + \varepsilon_i(\varepsilon\, d_j + g\, t - d_i), \tag{5}$$

where ε_j, d_j are the sign and digit of h_j.

The neighbor graph is the minimal set of neighbor types closed under refinement (5) that is reachable from the identity. A vertex σ is an *actual neighbor* if and only if the directed graph of children (5) admits an infinite path starting from σ [8].

3.2. **Invariant bounding set.**

Lemma 6 (Bounding rectangle)**.** *For every $n \geq 3$, the attractor satisfies $A_n \subset R$, where*

$$R = [0,\, 2n{-}3] \times [0,\, n{-}1{-}1/n]. \tag{6}$$

Proof. Write $R_0 = [0, 2n{-}3]^2$. A direct computation shows $\bigcup_j f_j(R_0) \subset R_0$, so $A \subset R_0$ by Hutchinson's theorem. Applying one Hutchinson step to R_0:

- Each spine piece $f_j(R_0)$, $0 \leq j \leq n{-}2$, satisfies $x_2 \leq \big(2n{-}3{+}(n{-}2)(n{-}1)\big)/n = (n^2{-}3n{+}5)/n$.
- The tooth piece $f_{n-1}(R_0)$ satisfies $x_2 \leq (n^2{-}n{-}1)/n = n{-}1{-}1/n$.

Since $(n^2{-}n{-}1)/n \geq (n^2{-}3n{+}5)/n$ for $n \geq 3$, the overall x_2-range of $\bigcup f_j(R_0)$ is $[0,\, n{-}1{-}1/n]$. The x_1-range remains $[0, 2n{-}3]$ (unchanged by the spine and tooth strips). A second Hutchinson step verifies that $R = [0, 2n{-}3] \times [0, n{-}1{-}1/n]$ is invariant. □

We also record the *refined bound* $A \subset B_1 = \bigcup_j f_j(R)$, which decomposes into rectangular strips:

$$\begin{aligned} &\text{spine strip:} \quad x_1 \in [0,\, n{-}1{-}1/n], \quad x_2 \in [0,\, (n^2{-}3n{+}5)/n], \\ &\text{tooth strip:} \quad x_1 \in [n{-}2{+}1/n,\, 2n{-}3], \quad x_2 \in [(n{-}1)(n{-}2)/n,\, n{-}1{-}1/n]. \end{aligned} \tag{7}$$

In particular, points with $x_1 \geq n{-}1$ lie exclusively in the tooth strip.

3.3. **Level-1 neighbors.**

Proposition 7 (Level-1 neighbors)**.** *The children of the identity under* (5) *that satisfy $A \cap \sigma(A) \neq \emptyset$ are precisely:*

$$\begin{aligned} \sigma_1 &= (+I,\ (n{-}1,\, 0)), && \textit{spine, forward}, \\ \sigma_2 &= (+I,\ (-(n{-}1),\, 0)), && \textit{spine, backward}, \\ \sigma_3 &= (-I,\ (3n{-}4,\, 2n{-}3)), && \textit{spine–tooth, far}, \\ \sigma_4 &= (-I,\ (2n{-}3,\, 2n{-}3)), && \textit{spine–tooth, near}. \end{aligned}$$

Proof. The children of the identity are $\sigma_{ij} = f_i^{-1} f_j$ with $i \neq j$. By (5), $\sigma_{ij}(x) = \varepsilon_i \varepsilon_j\, x + \varepsilon_i (d_j - d_i)$.

Spine–spine ($i, j \leq n{-}2$): the sign is $+I$ and the translation is $((j{-}i)(n{-}1),\, 0)$. For overlap, $A \cap (A{+}t) \neq \emptyset$ requires $|t_1| \leq \mathrm{width}_{x_1}(A) \leq 2n{-}3$. Thus $|j{-}i|\,(n{-}1) \leq 2n{-}3$, giving $|j{-}i| \leq 1$ (since $(2n{-}3)/(n{-}1) = 2 - 1/(n{-}1) < 2$). This yields σ_1 and σ_2.

Spine–tooth ($i \leq n{-}2,\ j = n{-}1$ or vice versa): the sign is $-I$ and the translation is $d_{n-1} - d_i = \big((n{-}1)(n{-}i) - 1,\ 2n{-}3\big)$. For overlap with sign $-I$, the translation must lie in $A + A \subset [0,\, 2(2n{-}3)] \times [0,\, 2(n{-}1{-}1/n)]$. The x_1-constraint gives $(n{-}1)(n{-}i) - 1 \leq 4n{-}6$, hence $n{-}i \leq (4n{-}5)/(n{-}1) < 4$, so $i \geq n{-}3$. This yields σ_3 (from $i = n{-}3$) and σ_4 (from $i = n{-}2$).

All other candidates are infeasible by the R-bound. □

3.4. **Level-2 neighbors and graph closure.**

Theorem 8 (Neighbor graph)**.** *For every integer $n \geq 3$, the neighbor graph of the comb IFS consists of exactly 6 neighbor types $\sigma_1, \ldots, \sigma_6$, where $\sigma_1, \ldots, \sigma_4$ are as above and*

$$\sigma_5 = (-I,\ (2n{-}3,\ n{-}2)), \qquad \textit{level-2, diagonal},$$
$$\sigma_6 = (-I,\ (n{-}2,\ n{-}2)), \qquad \textit{level-2, reflected}.$$

In particular, the Open Set Condition holds for all $n \geq 3$ (and trivially for $n = 2$, where A_2 is a rectangle).

Proof. We compute the children (5) of $\sigma_1, \ldots, \sigma_4$, identify the new types σ_5, σ_6, verify graph closure, and eliminate one spurious candidate τ.

Step 1: Children of σ_1 and σ_2. For $\sigma_1 = (+I, (n{-}1, 0))$, the "$gt$" vector is $g(n{-}1, 0) = (0, n{-}1)$. The spine–spine children have sign $+I$ and translation $((k{-}j)(n{-}1),\ n{-}1)$. Since $n{-}1 > n{-}1{-}1/n = \text{height}(R)$, the x_2-component exceeds the diameter of $A{-}A$ in the x_2-direction: all spine–spine children are infeasible.

The only surviving child comes from the tooth–spine case $(i, j) = (n{-}1,\ n{-}2)$, which produces $\sigma_5 = (-I, (2n{-}3, n{-}2))$ via the map f_{n-1}. By symmetry, σ_2 also produces σ_5 (via f_{n-2}).

Step 2: Children of σ_3. The only surviving child is from $(i, j) = (n{-}1, n{-}1)$ (tooth–tooth), producing $\sigma_6 = (-I, (n{-}2, n{-}2))$ via f_{n-1}.

Step 3: Children of σ_4. The surviving children are:

- $(i, j) = (n{-}2,\ n{-}1)$: σ_1, via f_{n-2};
- $(i, j) = (n{-}1,\ n{-}2)$: σ_2, via f_{n-1};
- $(i, j) = (n{-}2,\ n{-}2)$: σ_3, via f_{n-2};
- $(i, j) = (n{-}1,\ n{-}1)$: $\tau = (-I, (n{-}2, 2n{-}3))$, via f_{n-1}.

The candidate τ is not among the six types; we handle it below.

Step 4: Children of σ_5. The spine–spine children $f_i^{-1}\sigma_5 f_j$ with $i, j \leq n{-}2$ have sign $-I$ and translation $\big((n{-}1)(n{-}1{-}i{-}j) - 1,\ 2n{-}3\big)$. Among these:

- $i + j = n{-}4$: produces σ_3 ($n{-}3$ ways, maps $f_0, \ldots, f_{n-4}$);
- $i + j = n{-}3$: produces σ_4 ($n{-}2$ ways, maps $f_0, \ldots, f_{n-3}$);
- $i + j = n{-}2$: produces τ ($n{-}1$ ways); handled below.

The tooth–spine child $(i, j) = (n{-}1,\ 0)$ gives σ_1 via f_{n-1}, and the spine–tooth child $(i, j) = (0,\ n{-}1)$ gives σ_2 via f_0. All other children are infeasible.

Step 5: Children of σ_6. The spine–spine children have sign $-I$ and translation $\big((n{-}1)(n{-}1{-}i{-}j) - 1,\ n{-}2\big)$:

- $i + j = n{-}3$: produces σ_5 ($n{-}2$ ways, maps $f_0, \ldots, f_{n-3}$);
- $i + j = n{-}2$: produces σ_6 ($n{-}1$ ways, maps $f_0, \ldots, f_{n-2}$).

All other children of σ_6 are infeasible (by the R-bound or the B_1 refinement).

Step 6: Elimination of $\tau = (-I, (n{-}2, 2n{-}3))$. The candidate τ passes the coarse bounding-box test, so we must refine further. Among all n^2 children of τ, only one is feasible by the R-bound: the spine–spine child $(i, j) = (n{-}2,\ n{-}2)$, which produces $\tau' = (-I, (3n{-}4,\ n{-}2))$.

We eliminate τ' using the refined bound B_1. For $A \cap \tau'(A) \neq \emptyset$ with sign $-I$, we need $p + q = (3n{-}4,\ n{-}2)$ for some $p, q \in A$. Since $p_1 + q_1 = 3n{-}4$ and

$\max(x_1) = 2n-3$, both $p_1 \geq n-1$ and $q_1 \geq n-1$. By (7), points with $x_1 \geq n-1$ lie in the *tooth strip*, where $x_2 \geq (n-1)(n-2)/n$. Therefore

$$p_2 + q_2 \;\geq\; \frac{2(n-1)(n-2)}{n},$$

but the target is $p_2 + q_2 = n-2$. The inequality $2(n-1)(n-2)/n \leq n-2$ simplifies to $2(n-1)/n \leq 1$, i.e., $n \leq 2$, contradicting $n \geq 3$. Thus τ' is infeasible, so τ has no infinite path in the refinement graph and is not an actual neighbor.

Conclusion. Since σ_5 and σ_6 produce only children from $\{\mathrm{id}, \sigma_1, \ldots, \sigma_6\}$ (plus the eliminated τ), the graph is closed. Every vertex σ_k admits an infinite path: the cycle $\sigma_1 \to \sigma_5 \to \sigma_4 \to \sigma_1$ (and analogously for the others) ensures $A \cap \sigma_k(A) \neq \emptyset$. The finiteness of the neighbor graph implies the OSC [8]. □

Remark 9. For $n = 3$ the general formulas specialise: the role of $f_{n-3} = f_0$ collapses with the first spine map, so σ_5 has no B_3 contributions. The substitution matrix (see below) handles this automatically via $n-3 = 0$.

3.5. **Boundary substitution rules.** The proof of Theorem 8 determines the boundary GIFS explicitly. Writing $B_k = A \cap \sigma_k(A)$ for the six boundary pieces and reading off which map f_i produces each child, we obtain the substitution rules (valid for all $n \geq 3$):

$$\begin{aligned}
B_1 &= f_{n-1}(B_5),\\
B_2 &= f_{n-2}(B_5),\\
B_3 &= f_{n-1}(B_6),\\
B_4 &= f_{n-2}(B_1) \cup f_{n-2}(B_3) \cup f_{n-1}(B_2),\\
B_5 &= f_0(B_2 \cup B_3 \cup B_4) \cup \bigcup_{j=1}^{n-4} f_j(B_3 \cup B_4) \cup f_{n-3}(B_4) \cup f_{n-1}(B_1),\\
B_6 &= \bigcup_{j=0}^{n-3} f_j(B_5 \cup B_6) \cup f_{n-2}(B_6).
\end{aligned} \tag{8}$$

The 6×6 substitution (count) matrix M_n, with $(M_n)_{jk}$ equal to the number of occurrences of B_k on the right-hand side of the rule for B_j, is:

$$M_n = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & n-3 & n-2 & 0 & 0 \\ 0 & 0 & 0 & 0 & n-2 & n-1 \end{pmatrix}. \tag{9}$$

4. Disk-Like Property

A self-similar tile A is called *disk-like* if it is homeomorphic to the closed unit disk. As noted in the Introduction, tiles with large $\dim_{\mathrm{H}}(\partial A)$ are generically non-disk-like, so achieving $\dim_{\mathrm{H}}(\partial A)$ close to 2 while maintaining the disk-like property is particularly challenging.

The following criterion, due to Bandt and Wang [2], characterises disk-like self-affine tiles in the plane.

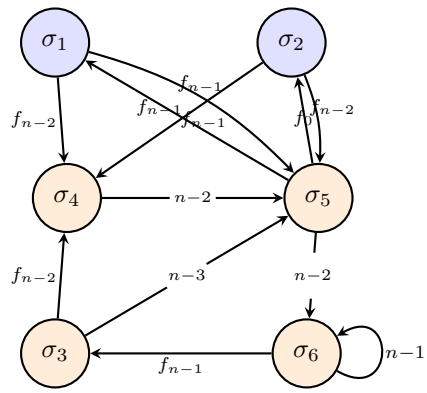

FIGURE 3. The boundary substitution graph of the comb IFS. Vertices are the 6 neighbor types: σ_1, σ_2 (spine–spine, sign $+I$; shaded blue) and $\sigma_3, \ldots, \sigma_6$ (spine–tooth or level-2, sign $-I$; shaded orange). A directed edge $\sigma_k \to \sigma_j$ means that the boundary piece B_k contributes to B_j under substitution (8). Edges of multiplicity 1 are labelled with the producing map f_i; edges of higher multiplicity show the count. The graph topology is independent of n (for $n \geq 3$); only the multiplicities depend on n.

Theorem 10 (Bandt–Wang [2])**.** *Let A be a self-affine tile in $\mathbb{R}^2$ whose associated tiling has finitely many neighbor types. Then A is disk-like if and only if, for every triple of distinct tiles $f_{i_1}(A)$, $f_{i_2}(A)$, $f_{i_3}(A)$ that pairwise intersect, the common intersection $f_{i_1}(A) \cap f_{i_2}(A) \cap f_{i_3}(A)$ consists of at most one point.*

We apply this criterion to the comb family.

Theorem 11 (Disk-like)**.** *For every integer $n \geq 2$, the comb tile A_n is disk-like.*

Proof. For $n = 2$, the attractor A_2 is a rectangle, so the claim is trivial. For $n \geq 3$, the proof has three steps: we identify the first-level intersection graph, show it has a unique triangle, and prove the corresponding triple intersection is a single point.

Step 1: First-level intersection graph. Two first-level tiles $f_i(A)$ and $f_j(A)$ intersect if and only if $f_i^{-1} \circ f_j$ is a neighbor. By Proposition 7, the only neighbors arising as children of the identity are $\sigma_1, \ldots, \sigma_4$. These correspond to the pairs:

- Spine consecutive: $(j,\, j{+}1)$ for $j = 0, \ldots, n{-}3$, giving σ_1 (or its inverse σ_2);
- Tooth: $(n{-}3,\, n{-}1)$ giving σ_3, and $(n{-}2,\, n{-}1)$ giving σ_4.

The resulting graph is a path $0{-}1{-}\cdots{-}(n{-}2)$ with vertex $n{-}1$ adjacent to $n{-}3$ and $n{-}2$. Its only clique of size 3 is $\{n{-}3,\ n{-}2,\ n{-}1\}$.

Step 2: Triple refinement is unique. Write $T = f_{n-3}(A) \cap f_{n-2}(A) \cap f_{n-1}(A)$ for the unique triple intersection. Substituting $A = \bigcup_k f_k(A)$,

$$T = \bigcup_{\substack{a,b,c \\ \text{pairwise} \\ \text{feasible}}} f_{n-3} \circ f_a(A) \ \cap \ f_{n-2} \circ f_b(A) \ \cap \ f_{n-1} \circ f_c(A),$$

where a triple (a, b, c) is *pairwise feasible* if $f_a^{-1}\sigma_1 f_b$, $f_b^{-1}\sigma_4 f_c$, $f_a^{-1}\sigma_3 f_c$ are all neighbors. Here $\sigma_1 = f_{n-3}^{-1} f_{n-2}$, $\sigma_4 = f_{n-2}^{-1} f_{n-1}$, $\sigma_3 = f_{n-3}^{-1} f_{n-1}$ are the three pairwise neighbor maps of the triangle.

The key observation: σ_1 and σ_2 each have exactly *one* surviving child under refinement (5). For $\sigma_1 = (+I, (n{-}1, 0))$, the vector $gt = g \cdot (n{-}1, 0) = (0,\, n{-}1)$. All spine–spine children $f_\alpha^{-1}\sigma_1 f_\beta$ (with $\alpha, \beta \le n{-}2$) have translation component $t_2' = n{-}1 > n{-}1{-}1/n = \mathrm{height}(R)$, so they exceed the R-bound and are infeasible. Among the remaining cases, a direct computation shows that the unique surviving child is

$$f_{n-1}^{-1} \circ \sigma_1 \circ f_{n-2} = \sigma_5,$$

which fixes $a = n{-}1$ and $b = n{-}2$.

With $a = n{-}1$ fixed, we evaluate the children of σ_3 at $(n{-}1,\, c)$. For spine indices $c \le n{-}2$, the translation has $|t_2'| = n{-}1 > \mathrm{height}(R)$ (infeasible), while for $c = n{-}1$ the child is

$$f_{n-1}^{-1} \circ \sigma_3 \circ f_{n-1} = \sigma_6.$$

This forces $c = n{-}1$. A consistency check confirms $f_{n-2}^{-1} \circ \sigma_4 \circ f_{n-1} = \sigma_1$. Thus the unique refinement triple is $(a, b, c) = (n{-}1, n{-}2, n{-}1)$, and the new pairwise state is $(\sigma_5,\, \sigma_1,\, \sigma_6)$.

Step 3: Period-2 cycle. Iterating the same analysis for the state $(\sigma_5,\, \sigma_1,\, \sigma_6)$:

- σ_1 (the "bottleneck") again forces $b = n{-}1$, $c = n{-}2$, giving σ_5.
- For σ_5, the child $f_\alpha^{-1}\sigma_5 f_{n-1}$ with spine α has translation $t_1' = -(n{-}1)(1{+}\alpha)$. Only $\alpha = 0$ yields $|t_1'| = n{-}1 \le 2n{-}3$, giving σ_2. So $a = 0$.
- $f_0^{-1} \circ \sigma_6 \circ f_{n-2} = \sigma_6$ (since $0 + (n{-}2) = n{-}2$).

The new state is $(\sigma_2,\, \sigma_5,\, \sigma_6)$.

One more iteration (using σ_2, which has the same bottleneck property as σ_1, giving a unique child σ_5 at $(a, b) = (n{-}2,\, n{-}1)$) returns to the state $(\sigma_5,\, \sigma_1,\, \sigma_6)$.

The refinement thus cycles with period 2:

$$(\sigma_5, \sigma_1, \sigma_6) \;\longrightarrow\; (\sigma_2, \sigma_5, \sigma_6) \;\longrightarrow\; (\sigma_5, \sigma_1, \sigma_6) \;\longrightarrow\; \cdots$$

At each level, the triple intersection is covered by a single level-k tile of diameter $O(n^{-k/2})$. As $k \to \infty$, the nested compact sets converge to a single point.

By Theorem 10, A_n is disk-like for all $n \ge 3$ (and hence for all $n \ge 2$). □

Remark 12. Both the neighbor graph (Theorem 8) and the disk-like property (Theorem 11) are proved for all $n \ge 3$ by the same structural argument: the combinatorial pattern of neighbors and their children depends only on the relative positions of the spine endpoint, the tooth, and their immediate predecessors, not on the spine length. Only the translations change with n.

5. Boundary Dimension

5.1. Derivation of the quartic.

Theorem 13 (Boundary dimension formula)**.** *For every integer $n \ge 3$, the characteristic polynomial of M_n factors as*

$$\det(\lambda I - M_n) = \lambda^2 \cdot p_n(\lambda), \tag{10}$$

where p_n is the quartic polynomial

$$p_n(x) = x^4 - (n{-}1)\, x^3 - 2\, x^2 - (n{-}1)(n{-}4)\, x + n(n{-}2). \tag{11}$$

The spectral radius $\lambda_n = \rho(M_n)$ is the unique root of p_n in the interval $(n{-}1,\, n)$, and the boundary Hausdorff dimension is

$$\dim_{\mathrm{H}}(\partial A_n) = \frac{2\log\lambda_n}{\log n}. \tag{12}$$

Proof. The dimension formula (12) follows from the Mauldin–Williams theory [9]: since all contractions f_j share the common ratio $r = 1/\sqrt{n}$ in the eigenspace of g, the Hausdorff dimension of the boundary GIFS equals $s = 2\log\rho(M_n)/\log n$.

We note that the OSC for the boundary GIFS (8) is inherited from the OSC of the original IFS: since the six boundary pieces $B_k = A \cap \sigma_k(A)$ are subsets of A, and distinct first-level tiles $f_i(A)$ have disjoint interiors by the OSC, the images $f_i(B_k)$ appearing in (8) also have mutually disjoint interiors. The Mauldin–Williams formula therefore applies.

To prove the factorisation (10), we solve the eigenvector equation $M_n v = \lambda v$ for $\lambda \neq 0$. From the first three rows:

$$B_1 = B_2 = \frac{B_5}{\lambda}, \qquad B_3 = \frac{B_6}{\lambda}.$$

Substituting into row 6: $\lambda B_6 = (n{-}2)B_5 + (n{-}1)B_6$, whence

$$B_6 = \frac{(n{-}2)\,B_5}{\lambda - (n{-}1)}. \tag{13}$$

Substituting all of B_1, B_2, B_3, B_6 in terms of B_5 into row 4:

$$\lambda B_4 = \frac{2}{\lambda}\,B_5 + \frac{(n{-}2)}{\lambda\big(\lambda-(n{-}1)\big)}\,B_5 = \frac{B_5}{\lambda}\;\cdot\;\frac{2\lambda - n}{\lambda-(n{-}1)},$$

giving

$$B_4 = \frac{(2\lambda - n)\,B_5}{\lambda^2\big(\lambda-(n{-}1)\big)}. \tag{14}$$

Finally, substituting into row 5:

$$\lambda B_5 = \frac{2}{\lambda}\,B_5 + \frac{(n{-}3)(n{-}2)}{\lambda\big(\lambda-(n{-}1)\big)}\,B_5 + \frac{(n{-}2)(2\lambda-n)}{\lambda^2\big(\lambda-(n{-}1)\big)}\,B_5.$$

Dividing by $B_5 \neq 0$ and multiplying through by $\lambda^2(\lambda-(n{-}1))$:

$$\begin{aligned}\lambda^3\big(\lambda-(n{-}1)\big) &= 2\lambda\big(\lambda-(n{-}1)\big)\\ &\quad + (n{-}3)(n{-}2)\lambda + (n{-}2)(2\lambda-n).\end{aligned}$$

Expanding and collecting terms, the right-hand side equals $2\lambda^2 + (n{-}1)(n{-}4)\lambda - n(n{-}2)$, so we obtain

$$\lambda^4 - (n{-}1)\lambda^3 - 2\lambda^2 - (n{-}1)(n{-}4)\lambda + n(n{-}2) = 0,$$

which is $p_n(\lambda) = 0$.

The factor λ^2 accounts for the two zero eigenvalues arising from the two linear dependencies in M_n: the identical rows for B_1 and B_2, and the relation $(n{-}2)\cdot(\text{row } B_1) + (n{-}1)\cdot(\text{row } B_3) = \text{row } B_6$. □

Remark 14 (Root location). The polynomial p_n satisfies:

$$\begin{aligned} p_n(n{-}1) &= -(n{-}2)(n^2{-}3n{+}1) &&< 0 \quad (n \geq 3),\\ p_n(n) &= 2n(2n{-}3) &&> 0 \quad (n \geq 3),\end{aligned}$$

so the intermediate value theorem guarantees a root in $(n-1,\, n)$. Since $p_n(0) = n(n-2) > 0$ for $n \geq 3$ and $p_n(1) = 2(n-2) > 0$, there is also a root between 1 and $n-1$; the dominant root is the one in $(n-1,\, n)$.

5.2. **Computed dimensions.** Table 1 lists the boundary dimension for selected values of n.

TABLE 1. Boundary dimension of the comb tile A_n for selected n. The polynomial is $p_n(x)$ from (11).

n	λ_n	$\dim_{\mathrm{H}}(\partial A_n)$	$2 - \dim_{\mathrm{H}}(\partial A_n)$
3	2.2213	1.4529	0.5471
4	3.3846	1.7590	0.2410
5	4.4790	1.8633	0.1367
6	5.5451	1.9120	0.0880
7	6.5951	1.9388	0.0612
8	7.6345	1.9550	0.0450
9	8.6665	1.9656	0.0344
10	9.6932	1.9729	0.0271
20	19.8279	1.9942	0.0058
50	49.9251	1.9992	7.67×10^{-4}
100	99.9613	1.9998	1.68×10^{-4}

5.3. **Root structure.** The polynomial $p_n(x) = x^4 - (n-1)\,x^3 - 2\,x^2 - (n-1)(n-4)\,x + n(n-2)$ has the following properties for $n \geq 3$:

(1) **Dominant root:** $\lambda_n \in (n-1,\, n)$, real and simple, with $\lambda_n = n - 4/n + O(1/n^2)$.
(2) **Second real root:** $\mu_n \in (1,\, n-1)$, with $\mu_n = 1 + 2/n - 14/n^3 + O(1/n^4)$ (note that the $1/n^2$ coefficient vanishes).
(3) **Two remaining roots α_n, β_n:** by Vieta's relations,

$$\alpha_n + \beta_n = (n-1) - \lambda_n - \mu_n = -2 + \frac{2}{n} + O(1/n^2),$$

$$\alpha_n \beta_n = \frac{n(n-2)}{\lambda_n\, \mu_n} = n - 4 + O(1/n).$$

The pair is complex conjugate with common modulus $|\alpha_n| = |\beta_n| = \sqrt{\alpha_n \beta_n} = \sqrt{n} - 2/\sqrt{n} + O(n^{-3/2})$.

Remark 15 (Coefficients as functions of n). The coefficients of p_n are polynomial in n:

$$p_n(x) = x^4 - (n-1)\,x^3 - 2\,x^2 - (n^2-5n+4)\,x + (n^2-2n).$$

In particular, the constant term $n^2 - 2n = n(n-2)$ is positive for $n \geq 3$, and the linear coefficient $-(n^2-5n+4) = -(n-1)(n-4)$ changes sign at $n = 4$. For $n < 4$, the linear coefficient is positive; for $n > 4$ it is negative. This sign change at $n = 4$ corresponds to a subtle shift in the boundary topology (although the neighbor graph retains 6 components throughout).

5.4. **Asymptotic expansion of the dominant root.**

Theorem 16 (Asymptotics)**.** *The dominant root λ_n of $p_n(x)$ satisfies*

$$\lambda_n = n - \frac{4}{n} + O\left(\frac{1}{n^2}\right) \qquad (n \to \infty). \tag{15}$$

Consequently,

$$2 - \dim_{\mathrm{H}}(\partial A_n) \sim \frac{8}{n^2 \ln n} \qquad (n \to \infty). \tag{16}$$

Proof. Step 1: Leading correction to λ_n. Write $x = n - u$ with $u = c_1/n + O(1/n^2)$. Substituting into $p_n(x) = 0$ and collecting powers of n, the dominant balance at order n^2 gives

$$p_n\big(n - c_1/n\big) = (-c_1 + 4)\, n^2 + O(n).$$

Setting the n^2 coefficient to zero yields $c_1 = 4$, hence $\lambda_n = n - 4/n + O(1/n^2)$.

Verification of the n^2 balance. We expand each term of $p_n(x)$ with $x = n - c/n$:

$$\begin{aligned} x^4 &= n^4 - 4cn^2 + 6c^2 + O(1/n^2), \\ (n{-}1)\, x^3 &= n^4 - n^3 + 3cn^2 - 3cn + \cdots, \\ 2\, x^2 &= 2n^2 - 4c + O(1/n^2), \\ (n{-}1)(n{-}4)\, x &= n^3 - 5n^2 + (4{-}c)\, n + O(1), \\ n(n{-}2) &= n^2 - 2n. \end{aligned}$$

The n^4 and n^3 terms cancel exactly; the n^2 coefficient is $-4c+3c-2+5+1 = -c+4$, confirming $c = 4$.

Step 2: From λ_n to $\dim_{\mathrm{H}}(\partial A_n)$. With $\lambda_n = n\big(1 - 4/n^2 + O(1/n^3)\big)$, we have

$$\log \lambda_n = \log n + \log\big(1 - 4/n^2 + O(1/n^3)\big) = \log n - \frac{4}{n^2} + O\left(\frac{1}{n^3}\right).$$

Therefore

$$\dim_{\mathrm{H}}(\partial A_n) = \frac{2 \log \lambda_n}{\log n} = 2 - \frac{8}{n^2 \log n} + O\left(\frac{1}{n^3 \log n}\right),$$

which gives (16). (Here log denotes the natural logarithm.) □

Remark 17 (Refined asymptotics). The next term in the expansion of λ_n can be extracted by the same method: write $x = n - 4/n + c_2/n^2$ and substitute into $p_n(x) = 0$. The n^2 coefficient was already forced to zero by $c_1 = 4$; balancing the coefficient of n (linear in c_2) gives $c_2 = 14$. Continuing to the next order yields $c_3 = -72$, so

$$\lambda_n = n - \frac{4}{n} + \frac{14}{n^2} - \frac{72}{n^3} + O(1/n^4).$$

Substituting into $\dim_{\mathrm{H}}(\partial A_n) = 2 \log \lambda_n / \log n$ and expanding, we obtain the refined gap formula

$$2 - \dim_{\mathrm{H}}(\partial A_n) = \frac{8}{n^2 \ln n}\left(1 - \frac{7}{2n} + O(1/n^2)\right).$$

5.5. **Numerical verification.** Table 2 compares the exact gap $2 - \dim_{\mathrm{H}}(\partial A_n)$ with the leading asymptotic $8/(n^2 \ln n)$. The ratio converges to 1 at the expected rate $1 - O(1/n)$.

TABLE 2. Convergence of $(2 - \dim_{\mathrm{H}}) \cdot n^2 \ln n/8 \to 1$.

n	$2 - \dim_{\mathrm{H}}(\partial A_n)$	$8/(n^2 \ln n)$	Ratio
10	2.706×10^{-2}	3.474×10^{-2}	0.779
20	5.769×10^{-3}	6.679×10^{-3}	0.864
50	7.666×10^{-4}	8.177×10^{-4}	0.937
100	1.680×10^{-4}	1.737×10^{-4}	0.967
500	5.114×10^{-6}	5.148×10^{-6}	0.993
1000	1.154×10^{-6}	1.158×10^{-6}	0.997
10000	8.683×10^{-9}	8.687×10^{-9}	1.000

5.6. **Spectral gap and monopole structure.** The substitution matrix M_n has characteristic polynomial $x^2 \cdot p_n(x)$, so its nonzero eigenvalues are exactly the four roots of the quartic.

Definition 18 (Spectral gap)**.** *Let λ_1 be the Perron eigenvalue of a boundary substitution matrix and $|\lambda_2|$ the second-largest eigenvalue modulus. The* spectral gap *is $\rho := |\lambda_2|/\lambda_1 \in [0, 1)$. We say the boundary GIFS is of* monopole type *if $\rho \ll 1$: the Perron eigenvector concentrates on a single component, meaning one boundary piece dominates the asymptotic iteration count while the remaining pieces contribute exponentially less (their shares decay as ρ^k after k substitution steps).*

For the comb family,

$$\rho_n = \frac{|\lambda_2|}{\lambda_n} = \frac{1}{\sqrt{n}} - \frac{2}{n^{3/2}} + O(n^{-5/2}) \longrightarrow 0,$$

so the boundary is of monopole type for every $n \geq 3$, with a spectral gap that improves monotonically.

The monopole structure is not a coincidence but a direct consequence of keeping the neighbor graph at constant size while driving $\dim_{\mathrm{H}}(\partial A_n) \to 2$. The boundary dimension satisfies $\dim_{\mathrm{H}}(\partial A_n) = 2 \log \lambda_n / \log n$, so approaching the ambient dimension requires $\lambda_n \to n$. But λ_n is the Perron eigenvalue of the 6×6 matrix M_n, whose trace equals $n-1$ (the only nonzero diagonal entry is $(M_n)_{66} = n-1$). Since the sum of all eigenvalues equals the trace, $\lambda_n \approx n$ forces the remaining roots to sum to approximately -1: they stay bounded while the dominant eigenvalue grows. In particular $|\lambda_2| = O(\sqrt{n})$, giving $\rho_n = O(1/\sqrt{n})$.

In other words, a *fixed-size* substitution matrix whose Perron eigenvalue grows without bound is automatically monopole. This explains why the combination of a small constant neighbor graph (6 pieces for all n) and high boundary dimension ($\to 2$) inevitably produces a strong spectral gap.

6. Discussion and Open Questions

(1) **Boundary dimension approaches 2 with constant graph.** The comb family provides the first explicit construction (to our knowledge) of an infinite parametric family of self-similar tiles with $\dim_{\mathrm{H}}(\partial A_n) \to 2$ and *bounded* neighbor graph complexity: exactly 6 components for all $n \geq 3$. This complements Keesling's examples [6], where $\dim_{\mathrm{H}}(\partial A)$ can be made arbitrarily close to 2 but the graph complexity grows without bound.

The rate $2 - \dim_{\mathrm{H}}(\partial A_n) \sim 8/(n^2 \ln n)$ is determined by the quartic polynomial $p_n(x)$; improving this rate would require a family whose boundary polynomial has a dominant root growing faster than linearly in n.

(2) **Disk-like property.** Theorem 11 proves the disk-like property for all $n \geq 3$ via the Bandt–Wang criterion: the triple intersection structure cycles with period 2 independently of n, mirroring the constancy of the neighbor graph (Theorem 8). Both results rely on the same structural observation: only the translations in the neighbor maps change with n, while the combinatorial pattern is universal.

(3) **Higher dimensions.** The comb construction generalises to $\mathbb{R}^d$ by using the companion matrix of $x^d - n$ and n maps with $n-1$ collinear translations and one off-axis map. The attractor has $\dim_{\mathrm{H}}(A) = d$ for every n, and we expect $\dim_{\mathrm{H}}(\partial A_n) \to d$ with bounded graph complexity. The boundary polynomial would then be determined by a family of polynomials parametrised by d and n.

(4) **Enumeration of disk-like tiles with high boundary dimension.** For each $n \geq 3$, classify all disk-like self-similar tiles with n maps whose boundary dimension exceeds $\dim_{\mathrm{H}}(\partial A_n)$. The comb tile may or may not be optimal within its class; a systematic search over all admissible orientation assignments could resolve this for small n.

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